QUANTUM COUPON COLLECTION: AN ALTERNATING SUM OF INVERSES

SUVRIT SRA*

ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: The matrix coupon-collector sum $\sum_{\emptyset\neq S}(-1)^{|S|-1}X_S^{-1}$ was conjectured positive definite for every order n and every dimension d. *Certificate:* Exact rational 3×3 matrices with $n=6$ and the integer vector $(4,1,3)^{\mathsf{T}}$ give a quadratic form strictly between -96 and -95.

Posed by S. Sra (2017); counterexample found by GPT Pro (2026-02-16); formalized by GPT-5.6 Pro, which re-derived the exact witnesses in 2026-08; audited by GPT-5.6 Pro; contributed by S. Sra.

In the classical coupon collector's problem with sampling probabilities $x_1,\dots,x_n>0$, the expected waiting time is the alternating sum

$$T_n(x_1,\dots,x_n)=\sum_{k=1}^{n}(-1)^{k+1}\sum_{1\le i_1<\cdots<i_k\le n}\frac{1}{x_{i_1}+\cdots+x_{i_k}},$$

whose positivity can be seen through a clever trick using the fact that $t\mapsto 1/t$ is completely monotone on $(0,\infty)$ (Flajolet et al., 1992). The statement quoted below asks whether the same holds with the positive numbers replaced by positive definite matrices.

Problem 1 (As posed: S. Sra, MathOverflow question 263833 (2017))**.** Here we take symmetric positive definite matrices $X_1,\dots,X_n\in\mathbf{S}^d_{++}$, and consider the sum

$$Q_n(X_1,\dots,X_n):=\sum_{k=1}^{n}(-1)^{k+1}\sum_{1\le i_1<\cdots<i_k\le n}(X_{i_1}+\cdots+X_{i_k})^{-1}.$$

Conjecture. For $n\ge 1$, and $X_1,\dots,X_n\in\mathbf{S}^d_{++}$, we have $Q_n(X_1,\dots,X_n)\succ 0$.
Note. The cases $n=1,2,3$ are trivially true. I've tried $n=4$ numerically, and it seems to hold. (Also, by CCP, positivity immediately holds for simultaneously diagonalizable matrices.)

The conjecture holds for a tuple of simultaneously diagonalizable matrices, where it is the classical identity coordinate by coordinate, and the source records that $n\le 3$ is trivial and that $n=4$ was checked numerically. Lemma 2 and Theorem 3 below upgrade that record: the conjecture is a theorem for every d and every $n\le 5$. It fails at $n=6$. Let us first present proofs of when it holds before a disproof of the general case.

Lemma 2 (Facet averaging)**.** *Let $A_1,\dots,A_m\in\mathbf{S}^d_{++}$, where $m\ge 2$, and put $A=\sum_{j=1}^m A_j$. Then*

$$A^{-1}\preceq\frac{m-1}{m^2}\sum_{j=1}^{m}(A-A_j)^{-1}.$$

Date: 1 September 2026.

* TU Munich, Dept. of Mathematics, School of CIT; Garching, Germany. Email: s.sra@tum.de.

From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Proof. Set $B_j = A - A_j$, which is positive definite because it is a sum of the remaining A_i. Since $\frac{1}{m}\sum_{j=1}^m B_j = \frac{m-1}{m}A$ and the inversion map is operator convex on $\mathbf{S}^d_{++}$,

$$\frac{m}{m-1}A^{-1} = \Big(\frac{1}{m}\sum_{j=1}^m B_j\Big)^{-1} \preceq \frac{1}{m}\sum_{j=1}^m B_j^{-1}.$$

Rearranging gives the claim. □

For a nonempty $S \subseteq [n]$ we abbreviate $X_S := \sum_{i\in S} X_i$; then, for fixed n, introduce the layer sums

$$E_k := \sum_{S\subseteq[n],|S|=k} X_S^{-1}, \qquad 1 \le k \le n,$$

each of which is positive definite. Applying Lemma 2 inside each $(k+1)$-subset T, with the A_j taken to be the X_i for $i \in T$, gives $X_T^{-1} \preceq \frac{k}{(k+1)^2}\sum_{i\in T} X_{T\setminus\{i\}}^{-1}$. Summing over all T of size $k+1$, and noting that each fixed k-subset arises as a facet of exactly $n-k$ of them, yields

$$E_{k+1} \preceq \frac{k(n-k)}{(k+1)^2}E_k, \qquad 1 \le k < n. \tag{1}$$

Theorem 3 ([**Proof**]: positivity through five variables)**.** *For every $d \ge 1$, every $1 \le n \le 5$, and every $X_1, \dots, X_n \in \mathbf{S}^d_{++}$, one has $Q_n(X_1, \dots, X_n) \succ 0$.*

Proof. $n = 1$ is immediate. Write $Q_n = E_1 - E_2 + E_3 - \cdots + (-1)^{n-1}E_n$ and apply (1) to obtain:

$$\begin{aligned}
n=2:&\quad E_2 \preceq \tfrac{1}{4}E_1, && Q_2 \succeq \tfrac{3}{4}E_1,\\
n=3:&\quad E_2 \preceq \tfrac{1}{2}E_1, && Q_3 \succeq \tfrac{1}{2}E_1 + E_3,\\
n=4:&\quad E_2 \preceq \tfrac{3}{4}E_1, \quad E_4 \preceq \tfrac{3}{16}E_3, && Q_4 \succeq \tfrac{1}{4}E_1 + \tfrac{13}{16}E_3,\\
n=5:&\quad E_2 \preceq E_1, \quad E_4 \preceq \tfrac{3}{8}E_3, && Q_5 \succeq \tfrac{5}{8}E_3 + E_5.
\end{aligned}$$

Each displayed lower bound is clearly positive definite. □

Theorem 4 ([**Disproof**]: quantum coupon collection)**.** *The conjecture is false: there are $d = 3$, $n = 6$ and $X_1, \dots, X_6 \in \mathbf{S}^3_{++}$ with $Q_6(X_1, \dots, X_6) \not\succeq 0$, so in particular $Q_6 \succ 0$ fails.*

Proof. Take $d = 3$, $n = 6$, $\varepsilon = 1/100$, and

$$X_i = w_i u_i u_i^{\mathsf{T}} + \varepsilon I_3,$$

where

i	u_i^{T}	w_i
1	$(-1, 2, 1)$	10
2	$(0, -3, 1)$	10
3	$(-3, -2, 0)$	100
4	$(2, -2, -2)$	100
5	$(-1, 0, 2)$	100
6	$(-1, 3, 0)$	100

Each X_i is strictly positive definite because $X_i \succeq \varepsilon I_3 \succ 0$, so the hypotheses are met exactly as posed. The example is genuinely noncommutative — the $(1, 2)$ entry of $[X_1, X_2]$ equals -1500 — and therefore lies outside the simultaneously diagonalizable regime in which the classical identity applies. For $v = (4, 1, 3)^{\mathsf{T}}$, exact rational evaluation of the sixty-three nonempty-subset terms gives

$$-96 < v^{\mathsf{T}} Q_6(X_1, \dots, X_6) v < -95.$$

The accompanying verifier performs this computation using only rational arithmetic. Thus, Q_6 has a negative quadratic form and is not positive semidefinite, let alone positive definite. □

The failure is not confined to the matrix ordering: the weaker scalar consequence $\operatorname{tr} Q_n > 0$ fails as well; *see Appendix A for more details and commentary.*

Appendix A. Some refinements of the quantum coupon collection conjecture

Theorem 4 refutes the conjecture in the Loewner order, which is what was asked. The failure is not confined to that ordering. Write

$$\Delta_n(X_1,\dots,X_n) := \operatorname{tr} Q_n(X_1,\dots,X_n) = \sum_{\emptyset\neq S\subseteq[n]} (-1)^{|S|-1} \operatorname{tr} X_S^{-1},$$

with $X_S := \sum_{i\in S} X_i$ as in the case. A positive definite matrix has positive trace, so the conjecture implies $\Delta_n > 0$, and Theorem 3 gives $\Delta_n > 0$ for every d and every $n \le 5$. This scalar consequence fails too, and for a reason one can see in closed form.

Let $u_1,\dots,u_n \in \mathbb{R}^3$ be such that every pair is linearly independent and every triple spans $\mathbb{R}^3$, and put

$$X_i(\varepsilon) := u_i u_i^{\mathsf{T}} + \varepsilon I_3, \qquad \varepsilon > 0,$$

so that $X_S(\varepsilon) = \sum_{i\in S} u_i u_i^{\mathsf{T}} + k\varepsilon I_3$ for $|S| = k$. A singleton leaves two regularized null directions, so $\operatorname{tr} X_S(\varepsilon)^{-1} = 2/\varepsilon + O(1)$; a pair leaves one, so $\operatorname{tr} X_S(\varepsilon)^{-1} = 1/(2\varepsilon) + O(1)$; and for $k \ge 3$ the unregularized matrix is already positive definite, so the trace of the inverse is $O(1)$. Only finitely many subsets occur, so the remainders sum uniformly and

$$\Delta_n(X_1(\varepsilon),\dots,X_n(\varepsilon)) = \frac{1}{\varepsilon}\Big(2n - \tfrac{1}{2}\binom{n}{2}\Big) + O(1) = \frac{n(9-n)}{4\varepsilon} + O(1). \tag{A.1}$$

The coefficient is negative exactly for $n \ge 10$, which the next theorem realizes with an exact witness.

Theorem A.1 ([**Disproof**]: the trace consequence)**.** *There are $d = 3$, $n = 10$ and $X_1,\dots,X_{10} \in \mathbf{S}^3_{++}$ with $\Delta_{10}(X_1,\dots,X_{10}) < 0$. A fortiori $Q_{10} \succ 0$ fails.*

Proof. Take $d = 3$, $n = 10$, $\varepsilon = 1/10000$, and, for $t = 0, 1, \dots, 9$,

$$u_t = (1, t, t^2)^{\mathsf{T}}, \qquad X_{t+1} = u_t u_t^{\mathsf{T}} + \varepsilon I_3.$$

Every X_i is strictly positive definite, and every three of the u_t are independent because their determinant is a nonzero Vandermonde product; the matrices are noncommuting, the $(1,2)$ entry of $[X_1, X_2]$ being 1. Exact rational evaluation of all $2^{10} - 1 = 1023$ nonempty-subset terms gives

$$-13901 < \Delta_{10}(X_1,\dots,X_{10}) < -13900.$$

The supplied verifier computes this with exact rational arithmetic and records the full numerator and denominator in `artifacts/certificate.json`. □

Remark A.2 (Scope). The record for the trace form is coarser than for the conjecture itself: true for $n \le 5$ by Theorem 3, false at $n = 10$, and undecided for $6 \le n \le 9$. Equation (A.1), whose coefficient vanishes at $n = 9$, says nothing there, since that construction is only one family; and a Q_n that is not positive semidefinite, such as the one of Theorem 4, can still have positive trace.

Code and certificate

The case is the directory `counterexamples/quantum-coupon-collector/` of the archive, at commit `8e31f6e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/quantum-coupon-collector
python verify.py
```

- Theorem 4 (result `quantum-coupon-collector`): exact arithmetic.

- `artifacts/certificate.json`: Exact rational n=6 Loewner witness, exact rational n=10 trace witness, certified sign intervals, and the structural checks (positive definiteness, noncommutativity, general position)

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT Pro (2026-02-16). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

ACKNOWLEDGEMENTS

The author (SS) thanks the Alexander von Humboldt (AvH) Foundation for their generous support via a Humboldt Professorship in AI.

REFERENCES

Philippe Flajolet, Danièle Gardy, and Loÿs Thimonier. Birthday paradox, coupon collectors, caching algorithms and self-organizing search. *Discrete Applied Mathematics*, 39(3):207–229, 1992. doi: 10.1016/0166-218X(92)90177-C.

Suvrit Sra. Quantum coupon collection: positivity of an alternating sum of matrices. MathOverflow question 263833, March 2017. URL https://mathoverflow.net/questions/263833/quantum-coupon-collection-positivity-of-an-alternating-sum-of-matrices. Posed 5 March 2017; unanswered as of 2 August 2026.